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A Study of High-Frequency Noise for Microplastics Classification Using Raman Spectroscopy and Machine Learning

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Abstract

Given the growing urge for plastic management and regulation in the world, recent studies have investigated the problem of plastic material identification for correct classification and disposal. Recent works have shown the potential of machine learning techniques for successful microplastics classification using Raman signals. Classification techniques from the machine learning area allow the identification of the type of microplastic from optical signals based on Raman spectroscopy. In this paper, we investigate the impact of high-frequency noise on the performance of related classification tasks. It is well-known that classification based on Raman is highly dependent on peak visibility, but it is also known that signal smoothing is a common step in the pre-processing of the measured signals. This raises a potential trade-off between high-frequency noise and peak preservation that depends on user-defined parameters. The results obtained in this work suggest that a linear discriminant analysis model cannot generalize properly in the presence of noisy signals, whereas an error-correcting output codes model is better suited to account for inherent noise. Moreover, principal components analysis (PCA) can become a must-do step for robust classification models, given its simplicity and natural smoothing capabilities. Our study on the high-frequency noise, the possible trade-off between pre-processing the high-frequency noise and the peak visibility, and the use of PCA as a noise reduction technique in addition to its dimensionality reduction functionality are the fundamental aspects of this work.

Keywords

Raman spectroscopy, microplastics, signal pre-processing, high-frequency noise, supervised classification, principal component analysis, PCA

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Introduction

Recent studies have shown evidence of the presence of microplastics in all kinds of bodies of water[1–5] which induces adverse effects on different environments, from marine ecosystems and biology[6–9] all the way to direct impact on human health.[10–13] In light of these studies, different organizations have striven to address this problem with new research projects, e.g., the Marie Skłodowska-Curie Innovative Training Networks (MSCA-ITN) Monitoring Micro- and Nanoplastics (MONPLAS) project,[14] and policies, e.g., the 2020 European Union Drinking Water Directive.[15]

Among the ordinary, yet necessary, tasks in this context is the identification of plastic materials, as there are several types of polymers labeled as "plastics". This topic has been addressed in the literature using different approaches such as microscopic, thermal, and spectroscopic methods.[16–18] Microscopic techniques can provide good morphological information, but obtaining information about the sample composition is almost impossible. Thermal methods can provide information about the sample composition but it is invasive and can sometimes be destructive and alter the original sample.

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In spectroscopy, on the other hand, light can be used to "fingerprint" materials by analyzing their detailed composition. In most cases, it is not necessary to touch the material; almost anything that can absorb, emit, or reflect light can be analyzed with a spectrometer.[19] For the analysis of plastic materials, spectroscopic approaches have been more reliable since they characterize the chemical structure of polymers.[20] The two most common methodologies for identification of microplastics are based on Fourier transform infrared spectroscopy (FT-IR) and Raman spectroscopy.

In a more specific manner, this work is focused on the application of Raman spectroscopy. Although the theory of light scattering was proposed in 1923 by Smekal, it was observed for the first time in experiments by Raman and Krishnan in 1928.[21] The interaction between monochromatic laser light and the molecules and atoms present in the polymer yields a frequency difference between the scattered light and the irradiating laser. This difference (often called Raman shift) can be detected and produces a material-specific spectrum.[22] The main advantages of working with Raman spectroscopy are the reliability for identification of materials (as the spectrum is unique), the ability to measure materials in a broad size range even larger than FT-IR,[23] and that it is non-invasive as it is non-destructive and contact-less.[24]

A Raman spectrum can be addressed as a high-dimensional data point, taking post-processing of this kind of data into the context of multivariate analysis. The main challenges present when working with Raman spectroscopic data are the high number of dimensions for each data point, as the number of spectral shifts used to obtain a relevant fingerprint of materials is usually above 1000, although it depends on the spectral bandwidth and resolution of the spectrometer. Additionally, the raw data obtained from the spectroscopic equipment have inherent noise of different kinds that must be corrected before applying any statistical/learning methodology.[25] Finally, the collected data highly depends on experimental conditions such as measurement setup, exposure time, laser power, objective, and beyond. All these facts imply additional challenges for data processing and material identification.

Particularly, the acquisition of Raman spectra leads to three main types of noise components: the fluorescence background (commonly known as baseline), the random cosmic spikes, and the high-frequency noise. This high-frequency noise originates from uncorrelated processes, such as detector/instrument noise, and it is often modeled as a Gaussian white noise process,[26] and reducing or removing it implies the application of signal-smoothing techniques.

Furthermore, the "fingerprint" extracted using Raman spectroscopy is mostly distinguishable for its peaks, which may be damaged by excessive smoothing on the collected sample. These facts open the question of a possible trade-off between pre-processing the high-frequency noise and the peak visibility.

In this paper, first, we systematically specify and detail a successful Raman signal pre-processing pipeline dedicated to microplastics, using carefully chosen techniques from the literature. Then, we investigate the impact of high-frequency noise on the performance of classification models for microplastic identification based on Raman spectroscopic signals. We thoroughly explore the advantages and disadvantages of using smoothing methods and dimensionality reduction techniques when high-frequency noise is present in the data. We show how signal smoothing, an essential step in the pre-processing of Raman signals, can be used to eliminate high-frequency noise, but it is clear that signal-smoothing techniques depend on user-defined parameters and it is impossible to tune the algorithms to work for all kinds of samples or experimental conditions. High-frequency noise and its pre-processing are important steps for the success of further classification modeling. We consider classification models that were successfully applied for the detection of microplastics[27] and we study in detail how high-frequency noise and signal smoothing can influence their performance. Furthermore, we show how principal component analysis (PCA) allows the construction of simpler and more robust models while barely sacrificing performance or adding extra overhead in the data classification time or resources, and it can also be used as a noise reduction technique. These fundamental aspects have not been investigated in the scientific literature dealing with the application of Raman spectroscopy and classification algorithms for microplastic detection.[27–31] The Appendix provided in the Supplemental Material provides details of the pre-processing algorithms used in this work. This study provides insight into pre-processing techniques for microplastic identification using Raman spectroscopy signals. Focus is given to the impact of high-frequency noise for the success of further classification modeling.

Materials and Methods

The methodology for the present work includes: (i) first, collecting the raw Raman signals; (ii) then applying the appropriate pre-processing steps to obtain relevant data that can be used as input for multivariate analysis. In the course of this paper, the data will be distinguished as "smooth" and "noisy." Smooth data means that the signal smoothing pre-processing step was applied, while noisy data means that the signal smoothing pre-processing step was not applied.

Once the data set is pre-processed, it is split into 60–20–20% training, validation, and test sets, respectively, which are ready for supervised classification tasks. Depending on the experiment, it can also include a PCA step to perform dimensionality reduction and compare the performance with the models in high dimensions.

A more general overview of this methodology is presented in the flowchart of Figure 1.

Sample Collection

All Raman signals used in this project were gathered using an inVia confocal Raman microscope using WiRE software, both

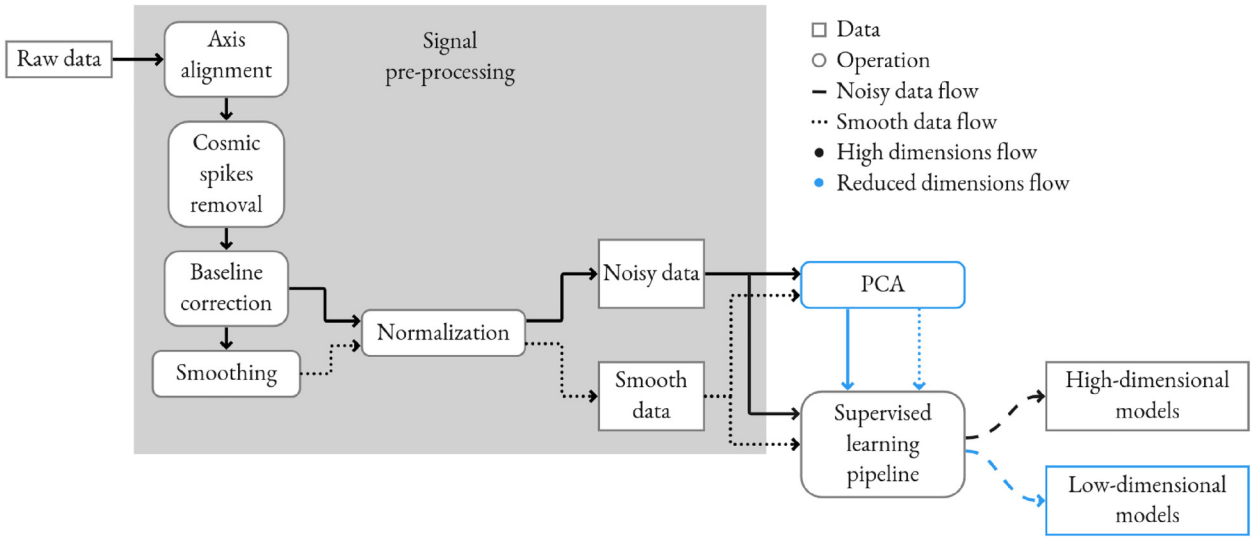


Figure 1. Flowchart of the general methodology followed. This chart shows the flow of all data signals collected, from the raw spectra until the resulting models are trained. The data first passes through a series of pre-processing steps (gray area), which bring the samples to a comparable scale and shape. The continuous lines represent the noisy (not smoothed) signals, and the dotted lines represent the smoothed signals (rectangular boxes at the end of the pre-processing). The samples are then used to train supervised classification models in both high dimensions (without PCA) and low dimensions (after the application of PCA, in light blue lines). This leads to classification models trained with smooth and noisy signals, which are used to evaluate the effect of high-frequency noise.

from Renishaw. Further details on the setup can already be found in the literature.[32]

Three main data sets were used, all containing Raman measurements of six types of microplastic samples[10]: polyethylene (PE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), and white polycarbonate (WPC). There were no pigments in the microplastic samples except for the latter, and they were all examined in a dry state as well as in water (wet) to study the Raman signal of microplastics in water suspension. However, it is well known that water does not affect the peak locations in a Raman spectrum. Thus, for simplicity, we decided to take the majority of the sample sets in dry conditions, and only some of them in wet conditions. This allowed us to have more diversity in our data while also allowing us to examine the effect of water on Raman signals.

The first two sets, namely `exposure_data` and `laser_data`, were, respectively, acquired with exposure times ranging from 0.01 to 8 s, and laser intensities between 0.0001% and 100% of the maximum, considering that the 532 nm laser has a maximum power of 50 mW and for the 785 nm laser, it is 100 mW. They were used to establish a baseline for the calibration of the pre-processing methods. One important remark is that these data sets were originally gathered to test the limits of the Raman microscope available at the lab. Although, they offer a wide variety of experimental conditions, the spectra recorded have anomalies beyond the expected in raw Raman data, which led to the construction of a third data set (`lab_data`) with newly collected samples in more restrained conditions. These conditions are specified in Table 1.

Table 1. Experimental conditions in `lab_data` for each material.

i	Laser power (mW)	Exposure time (s)	Condition	Magnification	No. samples
1	100	10	Dry	5×	10
2	10	10	Dry	5×	10
3	50	10	Dry	20×	5
4	1	15	Dry	5×	5
5	100	10	Dry	5×	10
6	5	15	Dry	5×	20
7	50	10	Dry	20×	10
8	100	10	Wet	5×	15
9	10	10	Dry	5×	10
10	1	15	Dry	20×	5

Data Pre-Processing

In Raman spectroscopy, spectra obtained from a spectrometer must be pre-processed before assessment. A raw spectrum is often altered by unwanted signals, such as fluorescence, detector noise, calibration errors, cosmic rays, and laser power fluctuations.[25] Additionally, raw data are processed to enhance important features that can be useful for further analysis. In general, a raw Raman signal is a composition of different components, but can be summarized as follows: let $s(\cdot)$ be the spectrum collected experimentally, then

$$s(\nu) = r(\nu) + b(\nu) + \xi(\nu) + \zeta(\nu), \quad \nu \in [\nu_1, \nu_2] \qquad (1)$$

where $r(\cdot)$ is the real Raman signal; $b(\cdot)$ is the baseline, fluorescence, background, or low-frequency noise inherent in Raman sampling; $\xi(\cdot)$ is a high-frequency uncorrelated (white) noise; $\zeta(\cdot)$ are the cosmic spikes/rays events signals that can randomly appear, especially for long exposure times; and ν is the associated Raman shift that moves on an interval $[\nu_1, \nu_2]$, which is either user-defined or limited by the equipment. One important remark is that $[\nu_1, \nu_2]$ is not a continuous interval in practice, but it is addressed as such for simplicity of notation. A better visualization of the composition of a raw Raman spectrum is presented in Figure S1 in the Supplemental Material.

The most common data pre-processing workflow consists of the following six steps, depending on the experimental conditions. These steps include spectral axis alignment, cosmic ray/spike removal, background/baseline correction, smoothing, normalization, and outlier removal, respectively. The workflow for pre-processing Raman signals was based on the protocols described in the literature.[16,25,26,33,34]

Spectral Axis Alignment. A Raman spectrum obtained from an experiment can be interpreted as a m-dimensional data point, where each of the m variables is a Raman shift (measured in cm^{-1}), and the actual value is the intensity (measured in arbitrary units). Formally, a Raman spectrum is a vector $\mathbf{r} = (\mathbf{r_1}, \ldots, \mathbf{r_m})^T \in \mathbb{R}^{\mathbf{m}}$.

In the present work, there were samples collected with different resolutions on the spectral axis, and, additionally, some samples were taken using only a sub-interval of all available shifts. In particular, linear interpolation was selected for axis alignment with constant extrapolation to 0 to replace the missing shifts.

Cosmic Spikes Removal. Cosmic spikes present in raw Raman spectra are due to high-energy particles passing through a charge-coupled device.[25] These devices are used in modern Raman instruments to increase sensitivity and reduce intrinsic noise.[35] These events are random and they produce a sharp spike in the raw spectrum (low width and high intensity). Therefore, cosmic spikes are considered outliers with no correlation with the data and they should be removed before any other pre-processing technique. The selected algorithm is the one proposed in,[35] due to its capability of only working with one sample individually. The pseudo-code of this algorithm can be found in the Appendix provided in the Supplemental Material. This procedure can be used to estimate $\zeta(\cdot)$ in Eq. 1.

Baseline Correction. This procedure is generally required due to fluorescence, which induces uneven intensity shifts across the spectrum.[25] The baseline correction procedure is often performed by applying differentiation in the signal data, polynomial fitting, or least squares schemes. The selected baseline correction algorithm is the iterative smoothing-splines with root error adjustment algorithm, proposed recently in Xu et al.[36] The general steps for this procedure are presented in the Appendix provided in the Supplemental Material. This procedure can be used to estimate $b(\cdot)$ in Eq. 1.

Signal Smoothing. This step is performed to eliminate the high-frequency noise present in the raw spectrum. In the field of signal analysis and processing, there are several studied techniques for smoothing a noisy signal. The most used methods for smoothing raw data[25] are moving average/median, Savitzky–Golay filtering,[37,38] and least squares.[39] This work selected a local regression approach: locally weighted scatterplot smoothing (LOWESS) originally proposed in Cleveland,[40] and the general algorithm to estimate the smoothed value is presented in the appendix provided in the Supplemental Material.

After having smoothed the pre-processed Raman signal (all the previous pre-processing steps have been executed), the Raman signal is normalized. This work uses the standard Euclidean norm per sample vector. The resulting Raman signal is a direct estimation of the real spectrum $r(\cdot)$ in Eq. 1 and can be now used for further analysis.

Supervised Classification of Raman Spectra

Recent reviews of artificial intelligence and machine learning methods for applications in Raman spectroscopy have been developed.[26,41,42] Almost all modern classification techniques from both multivariate analysis and machine learning have been applied for classification of material through Raman signals, these include artificial neural networks,[43–46] support vector machines,[47–49] tree-based methods,[50–52] Bayes-based methods[53–55] and discriminant analysis.[53,56,57] This work focuses on the application of two common classification methods for spectral data: support vector machines and linear discriminant analysis.[58,59]

Support Vector Machines and Error-Correcting Output Codes. A support vector machine (SVM) is an optimization-based (mainly) supervised classification model that attempts to create a hyperplane that better separates the boundaries of the classes involved, finding "support vectors" or points that will most likely be misclassified.[41] It has been widely studied and applied in the literature in different fields. The research found in literature[60–62] presents excellent introductions and developments of the theory behind SVMs. The main limitation of this model is that it can only learn to separate two classes at once and this paper deals with a six-class problem. The six classes represent the materials PE, PET, PP, PS, PVC, and WPC. One alternative to overcome this constraint is to use an error-correcting output code (ECOC) approach.

Error-correcting output code (ECOC) is a methodology to address multi-class classification problems using only binary, often called "weak" or partial, models.[63] This scheme was originally proposed by Dietterich and Bakiri[64] and has been

studied in the literature (see the literature[65–70]) with applications in different scientific areas (see the literature[63,71–73]). The general idea of ECOC for multi-class classification is presented in the Appendix provided in the Supplemental Material.

Linear Discriminant Analysis. Linear discriminant analysis (LDA) is a statistical classification technique that generalizes Fisher's discriminant work.[74] The intuitive idea underneath LDA is to find a projection of the data that is linearly separable by minimizing the inter-class variance and maximizing the distances between the means of the classes.[75] Similarly to SVM, LDA has been widely studied and applied in data mining and supervised learning. The introduction and theory for LDA classification can be found in the publications.[75–80]

Classification Performance

In the context of multi-class supervised classification, there always exists the challenge of measuring the overall performance of the classifier. Some methods have been proposed in the literature, for example, the metrics proposed by Carrillo et al.[81] and Kautz et al.[82] However, the implementation and understanding of these kinds of metrics are beyond the scope of the present work. Instead, a more classical approach was selected: standard and class-confusion matrices, and F1 scores.

(Class) Confusion Matrix. Confusion matrices are the most intuitive idea to measure classification performance: count the correctly and incorrectly classified points. Formally, for K classes, a confusion matrix is a matrix $\mathbf{C} \in \mathbb{Z}_{+}^{\mathbf{K}\times\mathbf{K}}$, where the entry c_{ij} is the count of samples classified to class j that actually belong to class i. Hence, the counts of correctly classified samples are on the diagonal of $\mathbf{C}$.

Once a confusion matrix $\mathbf{C}$ has been calculated, the concept can be extended for estimating K binary confusion matrices $\hat{\mathbf{C}}_k$ (here called class-confusion matrices), since these are key in the construction of binary metrics often used in classification performance. The main idea is to consider each class k as the positive class and the remaining $K-1$ classes as the joint-negative class.

F1 score. In the context of binary confusion matrices, the number of samples correctly classified as positives are called true positives (TPs), the correctly classified as negatives are called true negatives (TNs), the samples classified as positives that were actually negatives are called false positives (FPs), and finally, the count of samples predicted as negatives but were actually positives is called false negatives (FNs).

With the class-confusion matrices, the standard binary classification metrics can be applied for each class. In this manner, the selected metric is the F1 score: the harmonic mean between precision and recall. The precision is a measure of how precise is the model when classifying positive samples, as it is defined as the ratio between true positives and all samples labeled as positives. In contrast, the recall is the proportion of samples correctly classified as positive out of all the samples that are actually positive. Finally, the F1 score is a metric used to balance precision and recall, especially in the case where the number of TNsis large. A more detailed explanation of how to calculate this metric directly from the standard confusion matrix can be found in the Appendix provided in the Supplemental Material.

Results

All the code was developed in Matlab R2023a, and all results were obtained on a personal laptop running Arch Linux with an AMD Ryzen 5 4600H CPU and 16 GB of RAM.

Sample Collection and Pre-Processing

The number of samples per material and condition is presented in Table II for each of the data sets. The specific experimental conditions used for sample collection are presented for each material in Table I. The experimental conditions were varied across all collected data to promote diversity. In summary, `exposure_data` has 225 samples, `laser_data` has 271 samples and `lab_data` has 600 samples. In total, the joint data has 1096 samples which yield a data matrix $\mathbf{X} \in \mathbb{R}^{\mathbf{1096}\times\mathbf{3194}}$, where 3194 denotes the number of measured Raman shifts.

The data samples in `lab_data` for each material after applying all the pre-processing steps are presented in Figure 2. All samples have the same scale, they do not present baseline shift and show similar behavior overall despite being taken in different experimental conditions. This is proof of a successful pre-processing methodology.

Table II. Number of samples for all materials, distinguished by experimental conditions.

Data set	Condition	PE	PET	PP	PS	PVC	WPC
`Exposure_data`	dry	18	18	21	24	18	15
	wet	21	18	15	15	18	24
`Laser_data`	dry	30	21	24	24	47	30
	wet	12	21	12	16	22	12
`Lab_data`	dry	70	70	70	70	70	70
	wet	30	30	30	30	30	30

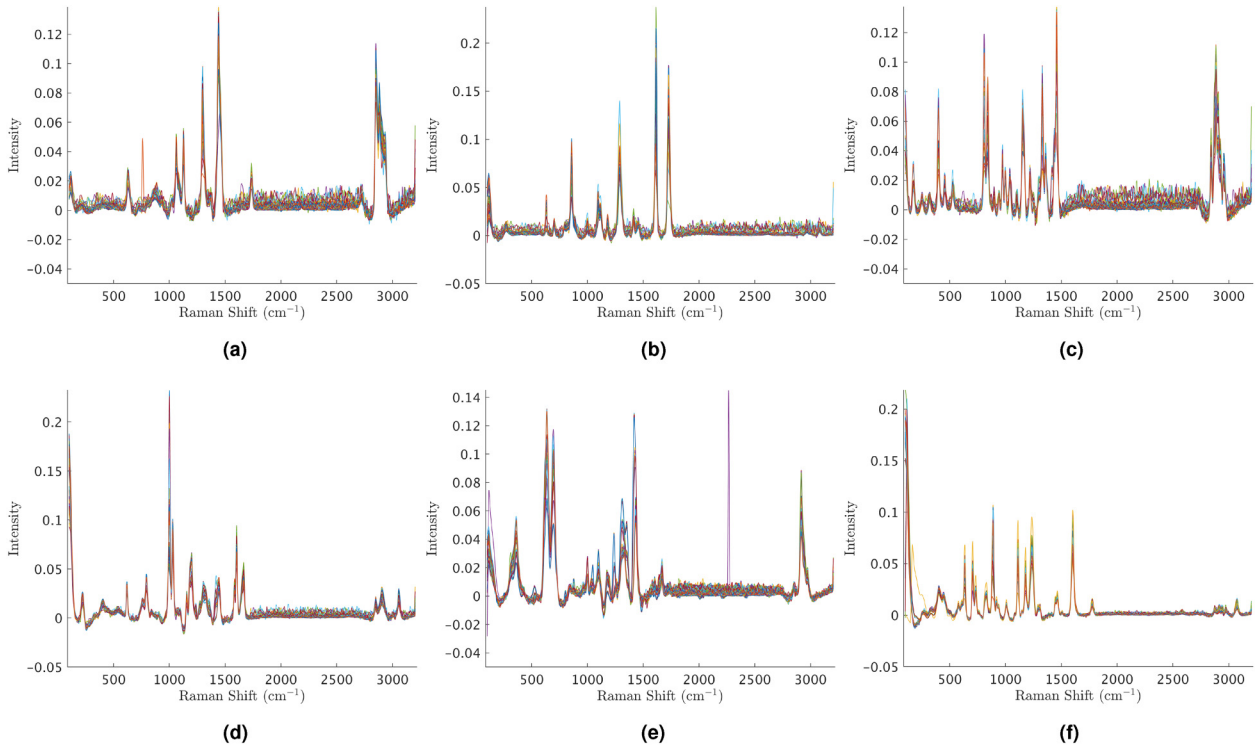


Figure 2. Raman spectra of all samples in `lab_data` after pre-processing: (a) PE; (b) PET; (c) PP; (d) PS; (e) PVC; and (f) WPC.

Influence of Wet Conditions. As discussed in Section Sample Collection, unlike FT-IR, it is possible to measure the Raman signal of microplastics immersed in water. In particular, within the short integration times required for capturing the Raman signal of polymers, no significant interference of water was observed.[83] The microplastics were immersed in deionized water only during the experiment to investigate the effect of water on the obtained Raman data. Figure S2 in the Supplemental Material shows the spectra of PE and PVC in both dry and wet conditions. Note how only the intensity of the peaks is affected by the experimental conditions, as expected. Moreover, the normalization of Raman signals in the pre-processing pipeline eliminates the differences in peak height, from which we can conclude that adding water to the samples we used for training will have little effect on the resulting signal. Hence, we only added a few wet samples for the diversification of the collected samples.

Microplastics Classification

Classification Without PCA. After applying the necessary pre-processing steps, the Raman signals from all data sets can be summarized into a matrix $\mathbf{X} \in \mathbb{R}^{1096\times3194}$ which can be used to train supervised classification models. The first set of models was trained using the pre-processed data, including the LOWESS signal smoothing, and tested with the validation, test, and "noisy" set, the latter being all samples collected with their original high-frequency noise. The goal is to compare the performance of the same models when different kinds of samples are introduced and gain insight into the impact of the high-frequency noise. Through the following results, we refer to "smooth" models as the models trained using data after signal smoothing.

The results were summarized using the confusion matrix of the test set and heat maps of F1 scores per set. The results for the ECOC with SVM learner and LDA are presented in Figure 3. We note that both ECOC (SVM) and LDA models perform well on all original sets, with a few samples misclassified and a high F1 score overall. The LDA model has high F1 scores for the training set and lower ones for the validation and test sets, this suggests slight overfitting and lower generalization capabilities when compared with the ECOC (SVM) model. Moreover, the ECOC model still excels in the classification of noisy samples, where all F1 scores were above 90%. On the other hand, the performance of the LDA model highly deteriorated: the maximum F1 score reached only 77% and some sets reached as low as 62%, which suggests that the LDA model cannot fully identify materials in the presence of high-frequency noise.In addition, to confirm the direct impact of high-frequency noise on the performance, the smooth LDA model was tested against smooth samples with synthetically added noise, with different standard

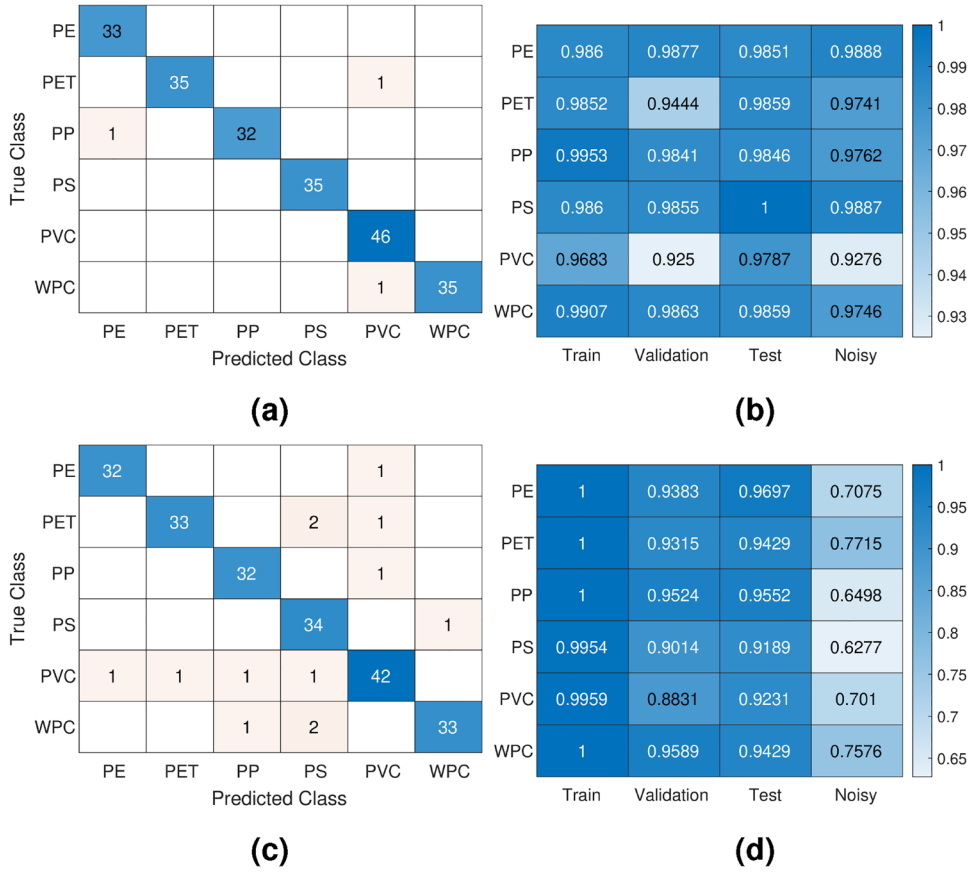


Figure 3. Classification results for smooth models on different sets for the models in high dimensions: (a) confusion matrix (test set) (ECOC); (b) heatmap of F1 scores (ECOC); (c) confusion matrix (test set) (LDA); and (d) heatmap of F1 scores (LDA).

deviations. Taking into consideration the composition of the raw sample in Eq. 1, it can be reorganized as

$$\xi(\nu) = s(\nu) - b(\nu) - r(\nu) - \zeta(\nu), \quad \nu \in [\nu_1, \nu_2] \qquad (2)$$

and, hence, the estimations $\hat{b}(\cdot)$, $\hat{r}(\cdot)$, and $\hat{\zeta}(\cdot)$ obtained from the pre-processing of the spectrum (see the previous section Data Pre-Processing), joined with the raw signal $s(\cdot)$, can be used to estimate the high-frequency noise $\xi(\cdot)$ (sometimes referred to as the residuals). From the residual signal, the standard deviation σ can be estimated and proposed a simple Gaussian model such that $\xi(\cdot) \sim N(0, \sigma^2)$ and adjust the noise for further experimentation. In Figure 4a, the histogram of all estimated σ is presented. Note that, roughly, it moves in the $S = [0,\ 0.5]$ interval.

In this manner, a sequence of different $\sigma \in S$ was generated and Gaussian noise was artificially added to each collected spectrum. Then, we used the smooth LDA model and plotted the obtained F1 scores for each material. This allowed us to investigate how the LDA model trained on smoothed samples performed on samples that might have not been properly smoothed. The results are presented in Figure 4b and show that the performance decreases hastily as the standard deviation increases for all materials. This is a clear evidence of the impact of high-frequency noise on LDA-like models.

Finally, an additional experiment was performed to display the impact of choosing a non-validated pre-processing technique. In this case, this was made by applying the LOWESS smoothing but varying the window size, these data were then used to feed the same smooth models, and the results can be observed in Figure 5. These plots show immediately the robustness of the ECOC (SVM) model over the LDA: (i) on the one hand, when the window size is close to zero, which means almost no smoothing, the ECOC classifies almost perfectly, while the LDA starts low, close on average to 50% of the F1 score; and (ii) on the other hand, for larger window sizes, the LDA model shows that the F1 scores decrease first, in terms of window sizes, than those of the ECOC approach.

Classification Including PCA. As briefly mentioned in the section Materials and Methods, PCA can be used for dimensionality reduction and it was used in this work for this aspect. Also, it was used as a noise reduction technique.[84] In this section, PCA was used before training the ECOC and LDA models, as it is a common methodology in multivariate analysis especially when dealing with spectroscopic data.[25]

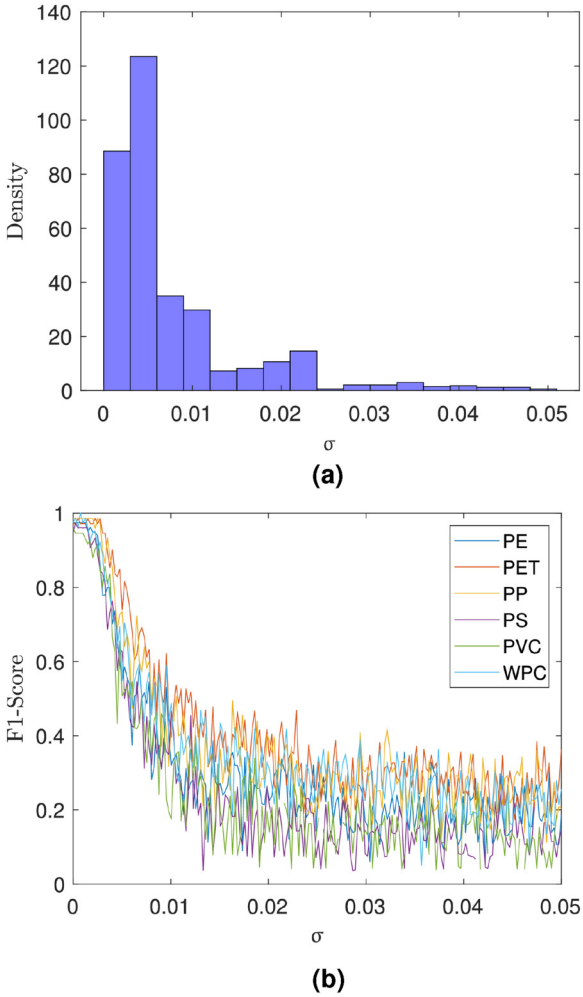


Figure 4. Classification results smooth LDA on samples with added noise: (a) distribution of σ for all collected samples; and (b) F1 score for smooth LDA on data sets with added noise ($\sigma \in S$).

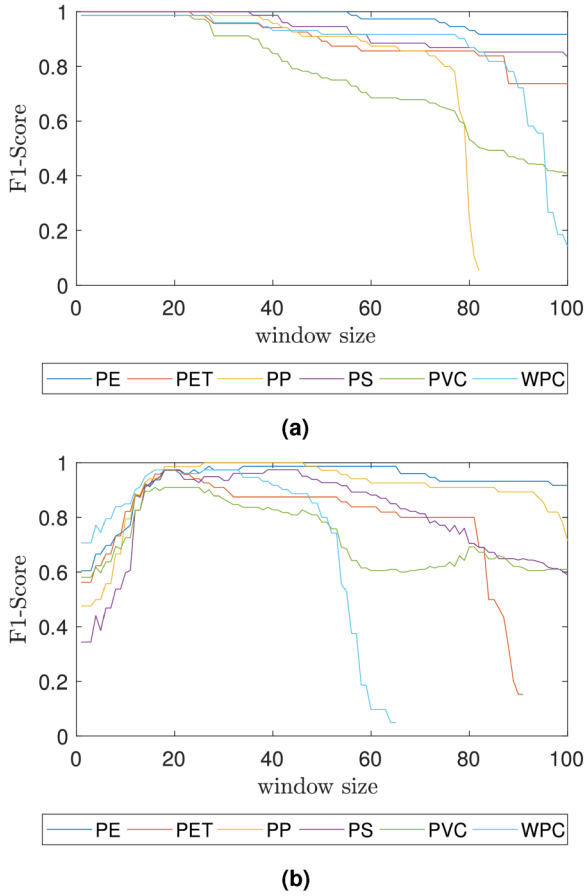


Figure 5. Classification performance for different LOWESS window sizes *h*: (a) ECOC (SVM) model and (b) LDA model.

Prinicipal component analysis (PCA) is applied to the data matrix **X** using the available Matlab function, with the singular value decomposition (default) option. There exist a lot of methods to select the number of dimensions to keep after applying PCA, such that the maximum amount of statistical information is retained within these dimensions. This work applies a more empirical approach using the "scree" plot (sorted eigenvalues λ of the sample covariance matrix), keeping the number of dimensions at the point where the plot breaks the steep slope. The scree plot for the data set is presented in Figure S3 in the Supplemental Material. It can be noted that around the 10th dimension, the plot breaks the slope and stabilizes close to zero; this means that the variance provided by the remaining dimensions is not as compelling as the first ten. Numerically, it can be proved that more than 90% of the data variance can be represented using the first 10 dimensions.

Same as for high dimensions, the first model was trained on smooth (reduced) samples and tested on the test, validation, and "noisy" sets. The results for the ECOC with SVM learner and LDA are presented in Figure 6. In this case, as opposed to the classifiers in high dimensions, the LDA outperforms the ECOC model with a slightly higher overall F1 score across all sets. In particular, for the LDA model, the noisy set is classified correctly in this case, with a minimum score of 94%. This is due to PCA serving as a noise reduction technique,[84] by eliminating redundant information.

Finally, this is also confirmed by plotting the first two components of the smooth and noisy data (see Figure S4 in the Supplemental Material document). From this plot, it was observed that the reduced data sets are almost identical in two dimensions, which already accounts for a high percentage of the total variance.

Discussion

This work successfully implemented and evaluated a preprocessing pipeline for microplastic classification, using relevant techniques extracted from the literature. Additionally, classification models previously proposed in the literature were implemented and applied to the Raman samples collected in various experimental conditions.

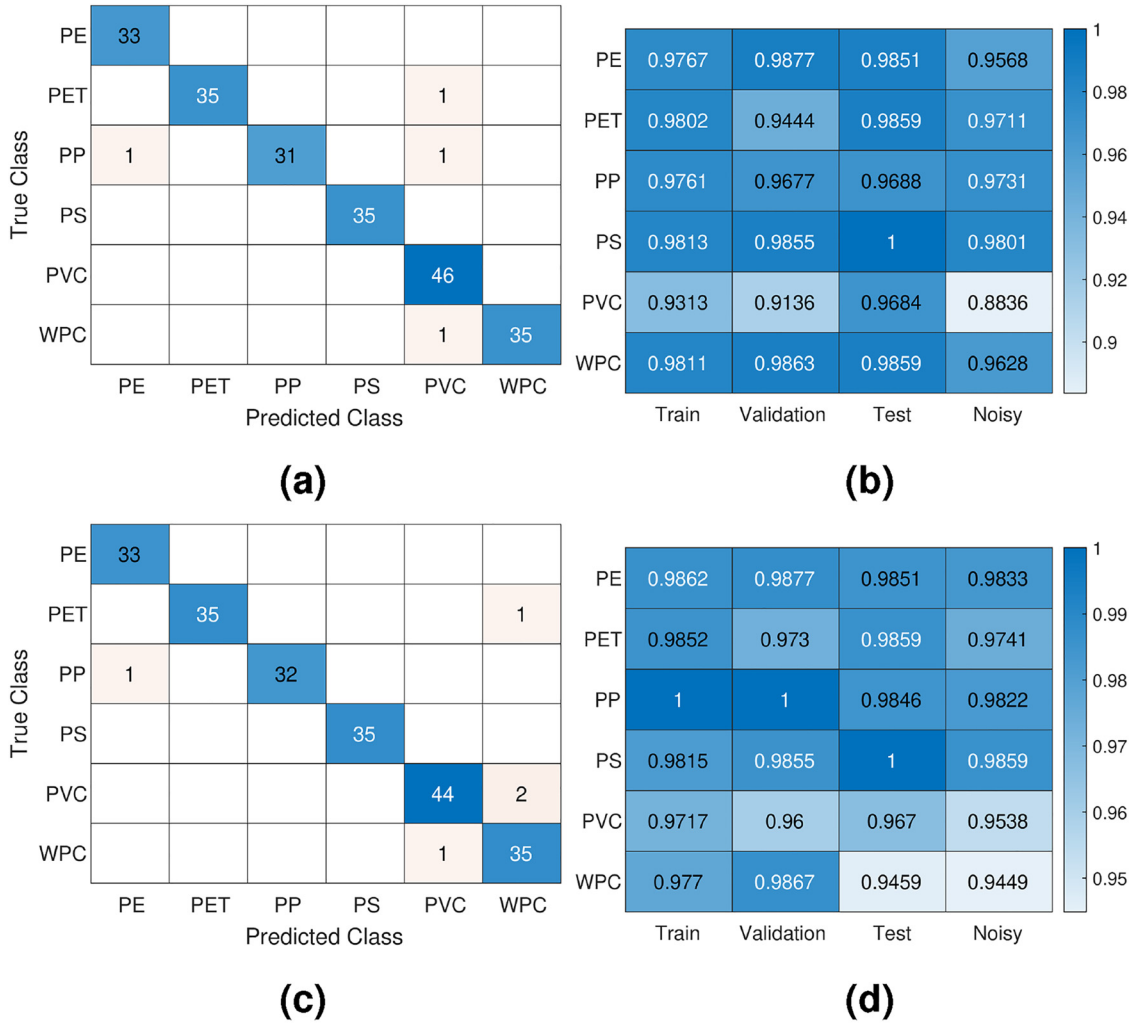


Figure 6. Classification results for smooth models on different sets (after PCA): (a) confusion matrix (test set) (ECOC); (b) heatmap of F1 scores (ECOC); and (c) confusion matrix (test set) (LDA).

Regarding the classification performance, one of the key findings was the larger generalization capabilities of the ECOC (SVM) model in comparison with a classic LDA approach on the high-dimensional signals. The experiment varying the LOWESS window size (Figure 5) summarizes this point well, where the ECOC (SVM) scores start extremely close to one with noisy signals and drop more for window sizes larger than those of the LDA model.

Additionally, from the noisy column in Figure 3d and the results depicted in Figure. 5, it can be concluded that the LDA model overfits the original training data, capturing the minimum details of the signal while the ECOC model captures a more general behavior in the training process. The former figure shows high F1 scores in the training data but not up to the same numbers in the noisy set, and the latter figure shows that the performance drops first in LDA compared to the ECOC (SVM). All models will eventually drop under this test since the signals become less relevant in terms of analytical value, i.e., peaks get less visible or destroyed by the smoothing. Furthermore, the application of PCA for dimensionality reduction can help to mitigate the issues raised with the smooth LDA model in high dimensions, when all obtained F1 scores are over 90% across all sets and for both classification models.

Conclusion

Signal smoothing is often referred to as an essential step in the pre-processing of Raman signals and in most cases, the high-frequency noise can be easily eliminated, but it is clear that signal-smoothing techniques depend on user-defined parameters and it is impossible to tune the algorithms to work for all kinds of samples or experimental conditions. In this sense, this work can conclude that an ECOC (SVM) is suggested when the smoothing procedure (or pre-processing in general) has not been carefully chosen and classification in high dimensions is required. Furthermore, PCA allows the construction of simpler and more robust models

while barely sacrificing performance or adding extra overhead in the data classification time or resources, and it can be used as a noise reduction technique (statistical smoother).

This detailed study on the advantages and disadvantages of using smoothing methods and dimensionality reduction techniques when high-frequency noise is present in Raman spectroscopic data could be of interest to other application domains where Raman signals are analyzed for classification tasks.

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Supplemental Material

All supplemental material mentioned in the text is available in the online version of the journal.

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